Altitudes, medians, and bisectors of triangles

1

a A median.

b A perpendicular bisector.

c An altitude.

d An altitude.

e An angle bisector.

f An angle bisector, a perpendicular bisector, an altitude, and a median.

2 $x = 46.5$

3 $x = 4$

4 $x = 8.5$

5

a $x = 9$

b $AD = 14$

6

a $x = 5$

b $BD = 12$

7

a $x = 32$

b $BD = 7$

8 The altitude $\overline{MQ}$ to $\triangle RMP$ formed two right triangles which are $\triangle RQM$ and $\triangle MQP$. Note that $\triangle RMP$ is isosceles with $\overline{RM} \cong \overline{MP}$ which are the hypotenuses of $\triangle RQM$ and $\triangle MQP$, respectively. Observe that the altitude $\overline{MQ}$ is a shared leg of two triangles. By Reflexive Property, $\overline{MQ} \cong \overline{MQ}$. By Hypotenuse-Leg (HL), $\triangle RQM \cong \triangle MQP$. With the use of CPCTC, $\angle RMQ \cong \angle QMP$. Consequently, $\overline{MQ}$ is an angle bisector of $\angle RMP$.

Additional questions

9

a An altitude is a line segment from the vertex of a triangle that is perpendicular to the opposite side

b It is an Orthocenter. In other words, it is a line segment joining a vertex to the midpoint of the opposite side, thus bisecting that side

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a Median

b Altitude

c Neither

d Altitude

e Both median and altitude

f Altitude

11

a Median

b Altitude

c Median and altitude, can also be called a perpendicular bisector

12 An explanation should include the following:

- A proof that the median of an isosceles triangle forms two congruent triangles
- A proof that the two congruent triangles must be right triangles, and therefore form right angles to the base
- A conclusion that the median satisfies the definition of an altitude

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